

Best of PA: Museum Data Analysis

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Project Description

For this project, we were tasked to create an application that showcases our idea of the “Best of PA”. After much deliberation, we decided to narrow our focus to Pennsylvania’s myriad museums. Using a dataset generated with the help of a Google Maps web scraper, we investigated Pennsylvania’s vibrant museum scene using machine learning algorithms to gain insights and achieve our objective of discovering the “Best Museum of PA”.

Our datasets obtained via the web scraper gave us invaluable information such as museum category, museum rating, and up to 100 user-written reviews per museum. We used these data points to perform sentiment analysis, clustering analysis, and correlation analysis.

In addition to the machine learning dimension of this project, we also worked to create a simple web-based application using FastAPI for the backend and Streamlit for the frontend.

Significance of the Project

This project is significant because it tackles many of the concepts that we have discussed throughout this semester, such as prediction models and NPs. It uses a reliable dataset containing museum entries in Google Maps to provide a service to users wishing to compare their qualities.

Sometimes, rating alone is not enough to tell the full story of a museum's value. Factors like the sensitivity of information displayed in a museum, an inconvenient location, or simply unfortunate circumstances on the day of visit can unfairly affect a museum's overall rating on Google Maps. Considering the fact that ratings are the primary consideration for people researching places to visit—and one of the main features used to guide search results—this makes some museums in PA receive less attention than they might otherwise deserve.

This is why we believe an application like ours—which uses machine learning to more accurately gauge the sentiment involved with museums by analyzing the language in reviews—provides a very valuable public good. There is undoubtedly a vast wealth of cultural information held in the museums, which gets disregarded in search results simply because of their baseline metrics. Our hope for this project is that smaller museums, museums in remote locations, and museums that may have unfairly received an unfavorable public reputation can be given a fairer chance.

Data Collection

Finding a dataset online to bring our idea to life initially proved difficult. We wanted to determine which were the best museums in PA, and determined that Google Maps would be our ideal source of data, since it contains possibly the largest publicly accessible aggregation of geographic data pertaining to museums and other establishments. Unfortunately, at first, we had no way of freely accessing Google Maps information.

Thankfully, we found the open-source Google Maps Scraper developed by Omkar Cloud and used it to compile our dataset. Using a search query for museums, heritage centers, historic societies, and other public educational establishments, the scraper generated two .csv files. These files became our bespoke datasets, created specifically to meet our demands. One contains the full list of detailed reviews for each location, while the other contains overviews of the museums' names, website links (if found or applicable), and the number of reviews.

The initial number of scraped locations was 914, and because each location might have multiple associated reviews, this created 62,916 lines of raw data in the full .csv file. However, some of these locations were incorrectly identified as museums, so this number was reduced to 564 after cleaning. Some of the metadata collected by the scraper includes the reviewer's name, visitation date, review creation date, a profile picture image link, etc. Below is a screenshot of some of the entries in the overview.csv file. The full versions of both .csv files are available in the GitHub repository in the "datasets" folder.

```

115 Name: The Academy of Natural Sciences of Drexel University
116 Link: No Link
117 Reviews: 3934 reviews
118
119 Name: Museum of Illusions Philadelphia
120 Link: No Link
121 Reviews: 3854 reviews
122
123 Name: Benjamin Franklin Museum
124 Link: No Link
125 Reviews: 1323 reviews",https://www.fi.edu/?utm_source=google&utm_medium=googlemybusiness&utm_campaign=2021,(215) 448-1200,,The Franklin Institut
126 ChIJ245kxzzuNIGR18efmji6tw4,The Waterfront,"Large outdoor mall on the river offering dozens of stores, a cinema, restaurants & nightlife venues.
127 Link: No Link
128 Reviews: 7650 reviews
129
130 Name: Station Square
131 Link: No Link
132 Reviews: 5046 reviews
133
134 Name: Wharton Square
135 Link: No Link
136 Reviews: 1975 reviews
137
138 Name: The Waterworks Shopping Center
139 Link: No Link
140 Reviews: 1313 reviews
141
142 Name: Amity Square At the Waterfront
143 Link: No Link
144 Reviews: 240 reviews",http://www.waterfrontpgh.com/,(412) 476-9157,1,The Waterfront (Owner),https://www.google.com/maps/contrib/1036009087532713
145 ChIJ8ZlC-InxNIgRDUaGB-UvoSg,Phipps Conservatory and Botanical Gardens,Schenley Park's horticulture hub features botanical gardens & a steel-&gl
146 Link: No Link
147 Reviews: 8631 reviews
148
149 Name: Pittsburgh Botanic Garden
150 Link: No Link
151 Reviews: 1688 reviews
152
153 Name: The Octopus Garden

```

Data Processing and Feature Engineering

Preprocessing in this project began with cleaning our raw datasets. For the overview dataset, we began by dropping rows that did not contain significant information. Then, we filtered out all non-museum institutions and dropped irrelevant feature columns. For the review dataset, we filtered out all non-English reviews and then selected the relevant feature columns. After cleaning both datasets, we merged them into one master dataset.

We then proceeded to natural language processing (NLP). This involved lowercasing, removing punctuation and numeric text using a regular expression, word tokenization using the `str.split()` method, stop-word filtering, part-of-speech tagging, and emergent lemmatization.

The specific functionalities of our project required their own requisite preprocessing steps. For clustering analysis, we used principal component analysis (PCA) to select the best features for a 2-D plot of our data and one-hot encoding to transform non-numeric data into numeric data. Sentiment analysis and weighted feature ranking required certain features in the dataset to be scaled using a min-max scaler.

Model Development

Several machine learning models were used in this project. Sentiment analysis was performed using the TextBlob module, taking in the processed review texts as input and returning polarity (how positive or negative a review is) and subjectivity (how neutral or biased a review is) scores.

Clustering analysis was performed using a k-Means algorithm with k set to 4 (this value was chosen after inspection using the Elbow Method). To train the k-Means model, we used a subset of the scaled dataset created for sentiment analysis (the features used were scaled average polarity, scaled museum rating score, and scaled number of reviews). We also decided to include the non-numeric “categories” feature. We transformed that data into a numeric form using one-hot encoding. In order to visualize the clusters, we needed to reduce the high dimensionality of the data. Using PCA, we reduced the data to the two principal components that explained most of the variance.

Lastly, for our correlation analysis, we trained a simple linear regression model on the museum rating and average polarity features and used it to determine which museums’ Google Maps ratings were least correlated with their sentiment score. This was achieved using the residuals metric in the output. Testing the linear regression model showed that it was not a very good model ($R^2 = 4.7\%$ and $MSE = 0.116$), but we deemed it sufficient for this basic task.

Showcase

Our model is used in our application through the backend. After our model is trained and undergoes analysis and clustering, the results are stored in a dictionary, which is then used in the frontend to display on the web application.

Top Museums:

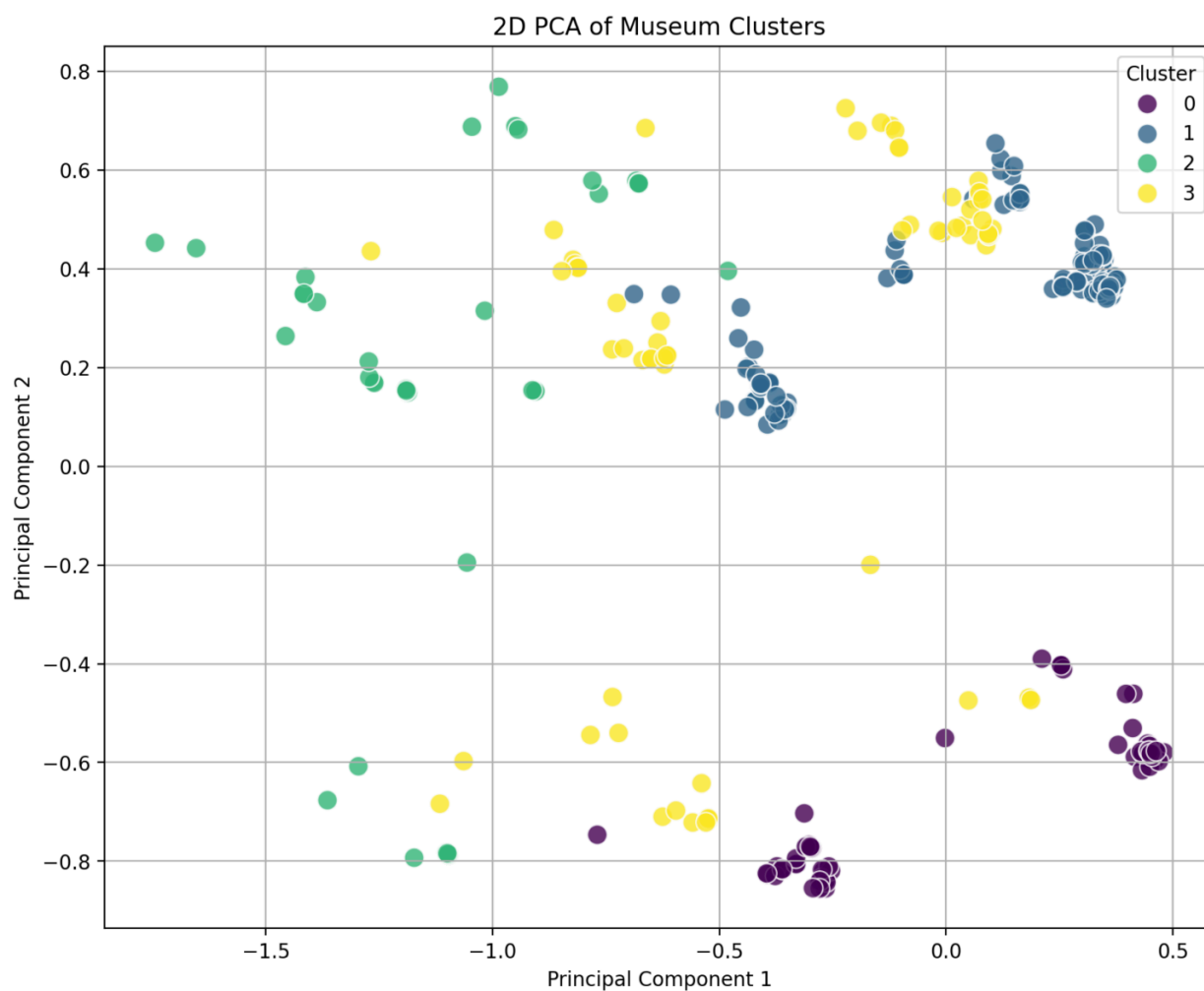
Top 10 Museums by Weighted Score

name	weighted_score
Philadelphia Museum of Art	0.7256
Frank Melega Art Museum	0.7
Women Veterans Museum	0.6153
Gettysburg National Military Park Museum & Visitor Center	0.6128
Beaver County Genealogy & History Center	0.6006
Philadelphia Museum of Art Steps	0.5932
The Franklin Institute	0.5907
Gather Place Museum	0.5889
Catlin House Museum	0.5731
Historic Catasauqua Preservation Association	0.56

Complete Museum Rankings

name	weighted_score
The Franklin Institute	0.5907
Gather Place Museum	0.5889
Catlin House Museum	0.5731
Historic Catasauqua Preservation Association	0.56
Titusville Historic Society	0.5582
Historical Society of Dauphin County	0.5519
Little Red Schoolhouse	0.5405
Grist Mill Dam	0.5312
American Industrial Mining Co. Museum	0.5298
Woodmere Maguire Hall	0.5297
Johnstown Children's Museum	0.5272

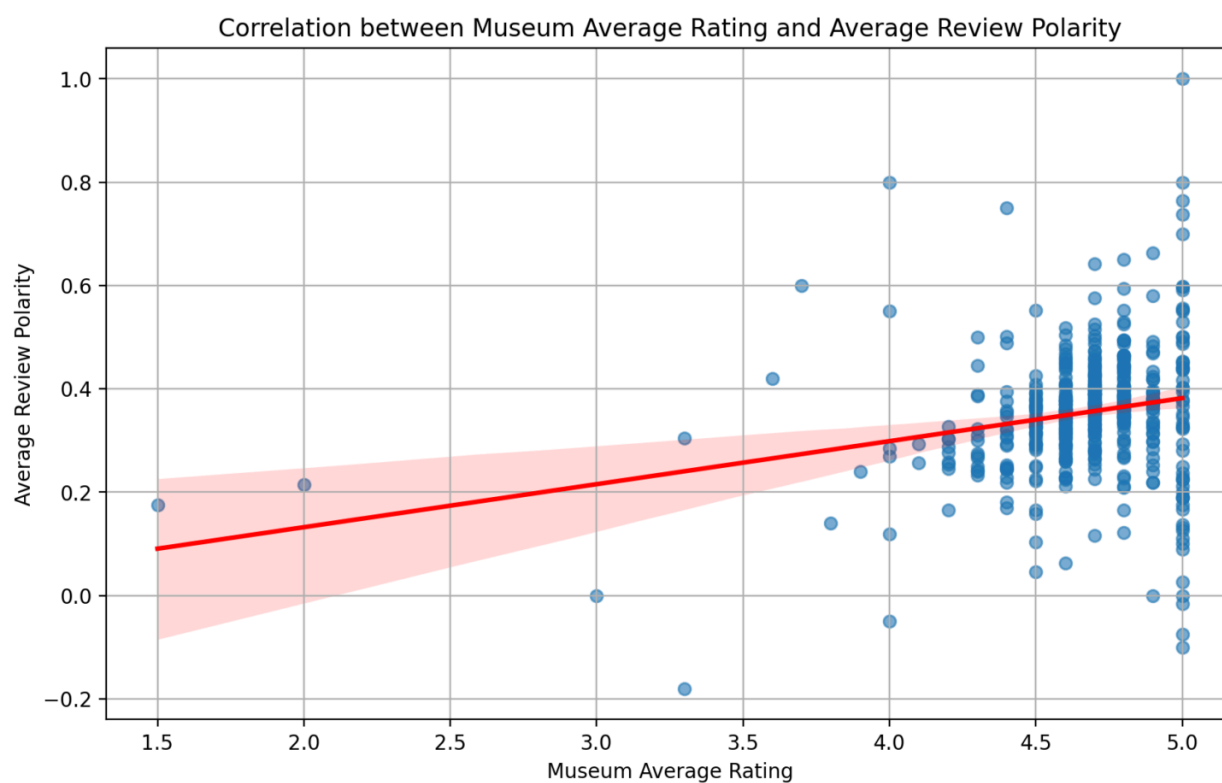
Clustering Analysis:



Cluster Summary Statistics

cluster	sc_average_polarity	sc_museum_average_rating	sc_number_of_reviews
0	0.4484	0.908	0.0102
1	0.4586	0.9043	0.0275
2	0.4502	0.903	0.0213
3	0.4484	0.8981	0.0432

Sentiment Correlation:



Museums with Lowest Sentiment-Rating Alignment

These museums have the biggest discrepancy between their ratings and the sentiment of their reviews.

name	museum_average_rating	average_polarity	residuals
Frank Melega Art Museum	5	1	0.618
Titusville Historic Society	4	0.8	0.5013
Conshohocken Historical Society	5	-0.1	-0.482
Hessian Powder Magazine Museum	5	-0.075	-0.457
Luzerne County Museum	3.3	-0.1812	-0.4216
Women Veterans Museum	5	0.8	0.418
Historic Catasauqua Preservation Association	4.4	0.75	0.418
Old York Rd Historical Society	5	-0.0167	-0.3986
Beaver County Genealogy & History Center	5	0.7649	0.3829
Lincoln on Gravity: The Lincoln Museum	5	0	-0.382

Cluster Details:

Cluster Characteristics

Top categories in each cluster:

Cluster 0	
Top 10 categories in this cluster:	
Category	Count
Museum	117
Art museum	11
Local history museum	9
Children's museum	7
Historical landmark	6
Rail museum	5
Event venue	5

Cluster 1	
Top 10 categories in this cluster:	
Category	Count
Tourist attraction	231
Museum	196
Art museum	35
Non-profit organization	20
Historical landmark	19
Park	11
Heritage museum	11
Local history museum	10
Historical society	10
Children's museum	9

Cluster 2

Top 10 categories in this cluster:

Category

Count

Historical place museum

49

Tourist attraction

38

Historical landmark

18

Museum

12

History museum

10

Local history museum

7

Historical place

6

Heritage museum

4

Garden

3

Park

3

Raw Data:

Raw Data Browser

Show all review data

Show museum summary data

All Reviews with Sentiment Analysis

Museum Summary

name	review_text	polarity	subjectivity	museum_average_rating
Philadelphia Museum of Art	beautiful museum excellent art piece	0.925	1	4.8
Philadelphia Museum of Art	visit new jersey side liberty state park total game changer crowd much thinner batter	0.2205	0.5786	4.8
Philadelphia Museum of Art	stop see famous step rocky run great experience	0.65	0.875	4.8
Philadelphia Museum of Art	recreate rocky	0	0	4.8
Philadelphia Museum of Art	spend day	0	0	4.8
Philadelphia Museum of Art	great place go run rocky step visit rocky souvenir shop	0.8	0.75	4.8
Philadelphia Museum of Art	visit philadelphia museum art short trip turn one memorable stop build impressive e	0.1328	0.5065	4.8
Philadelphia Museum of Art	big beautiful collection highly recommend	0.3367	0.5467	4.8
Philadelphia Museum of Art	must see tourist spot rocky course view night magnificent	1	1	4.8
Philadelphia Museum of Art	incredibly curated jaw drop masterpiece observe	0.9	0.9	4.8

name	museum_average_rating	average_polarity	review_rating
Philadelphia Museum of Art	4.8	0.3787	5
Gettysburg National Military Park Museum & Visitor Center	4.8	0.3523	3
Philadelphia Museum of Art Steps	4.7	0.3483	5
The Franklin Institute	4.6	0.3736	4
Crayola Experience	4.2	0.3273	5
Please Touch Museum	4.6	0.2844	5
The Mutter Museum at The College of Physicians of Philadelphia	4.6	0.2593	5
The Andy Warhol Museum	4.6	0.3509	5
Philadelphia's Magic Gardens	4.6	0.3154	5
The Amish Village	4.4	0.3767	5

Discussion and Conclusions

The findings shown above are the result of weighted feature ranking using sentiment analysis from the dataset we used with the help of the Google Maps Scraper. The Philadelphia Museum of Art ranked first with a score of 0.726, earning the title of “Best of PA”, followed closely by the Frank Melega Art Museum at 0.700. Other strong performers included the Women Veterans Museum, Gettysburg National Military Park Museum & Visitor Center, and The Franklin Institute. These top destinations have high ratings, sentiment, and number of reviews, which explains their strong performances.

In terms of our cluster summary, all four clusters had similar average weighted scores, ranging from about 0.448 to 0.459. This shows that the museum dataset is balanced overall, with no one cluster drastically outperforming any other. Average ratings were consistently high across clusters (around 4.5 out of 5), while sentiment averages remained strongly positive (around 0.90). The main difference between clusters comes more from variation in review sentiment spread rather than rating quality.

As for our sentiment correlation/analysis, museums that did not have written reviews that correlated with their rating were analysed here. Frank Melega Art Museum had the largest positive discrepancy, meaning visitors gave perfect ratings, but sentiment was less enthusiastic than expected. On the negative side, Conshohocken Historical Society and Hessian Powder Magazine Museum received high ratings despite noticeably weaker sentiment in review text, which may indicate polite scoring but mixed written feedback. Conversely, some museums, such

as the Women Veterans Museum and Beaver County Genealogy & History Center, showed stronger sentiment than ratings alone would predict.

The main issue we had with the project was finding the right dataset for our ideas, and it proved that no dataset on the internet included all the features we desired. Thus, finding the Google Maps Scraper program was the most pivotal moment during this project, as it enabled us to compile a dataset and use it to create our various models.

We tried to include as much of our course material as we could when creating our model. This included preprocessing techniques such as NLP and PCA (for clustering), and models such as k-Means and linear regression for our model. This project also helped us broaden our horizons as we researched and learned how to use the TextBlob module to get the polarity and subjectivity scores (sentiment) scores.

With more time and resources, we would be able to take this project even further by using geolocation data to place each museum in its county and perform additional clustering analysis by county. This would allow us to gain even more insights from our data and potentially crown the “PA County with the Best Museums”. Still, we are extremely proud of what we were able to accomplish given the time and resources we had, and we believe that we delivered an application that lives up to our objective of investigating and showcasing the “Best of PA”.

AI Usage

While no artificial intelligence (AI) tools were used to write any of the code for this project, we utilized AI tools to assist with basic debugging and troubleshooting and to provide advice on framework selection for both the backend and frontend of the web application. AI tools utilized include Google Gemini 2.5 and Anthropic Claude Haiku 4.5.

References

Omkar Cloud. (2026). *Google Maps Scraper* [Computer software]. Github.

<https://github.com/omkarcloud/google-maps-scraper>